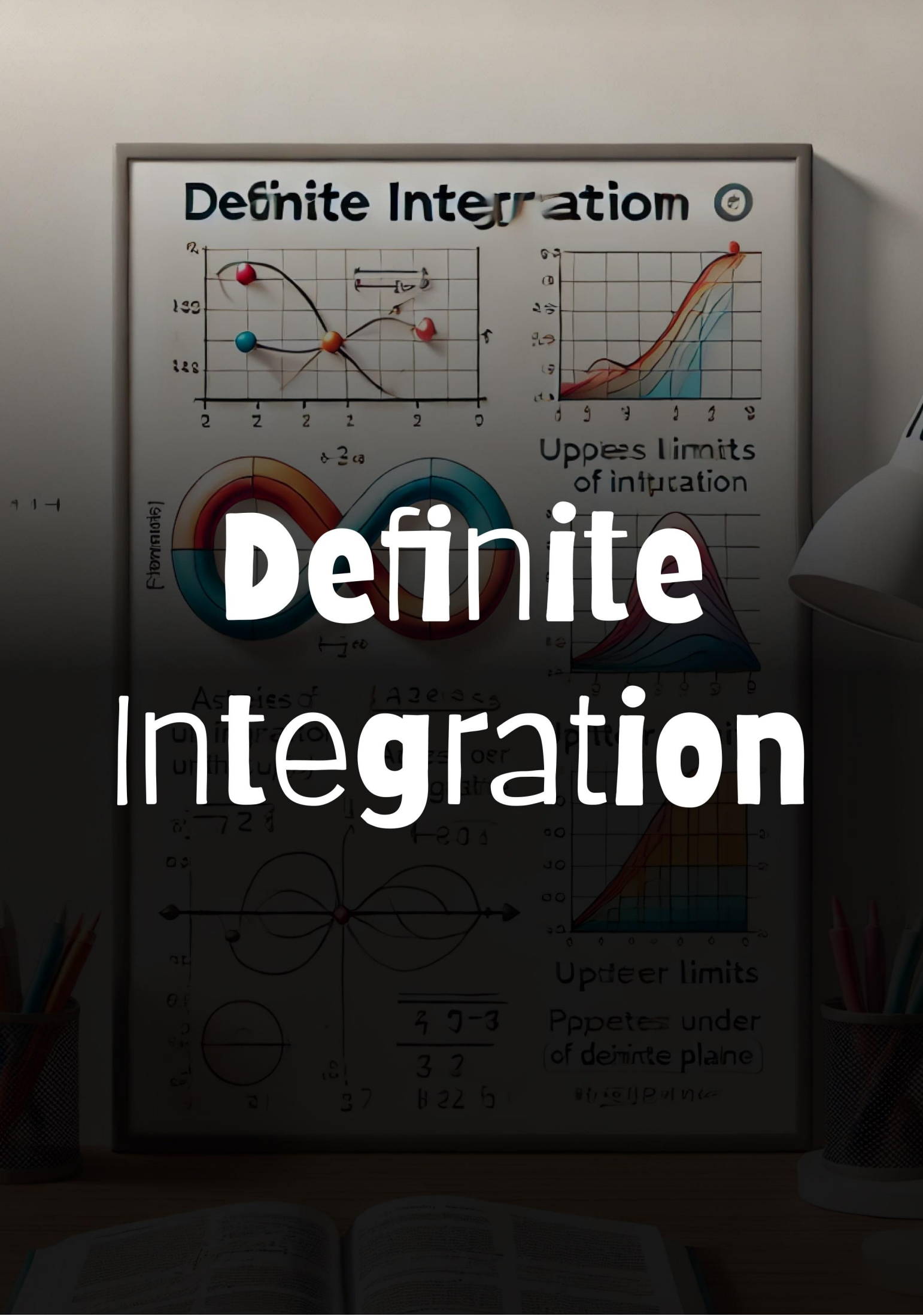
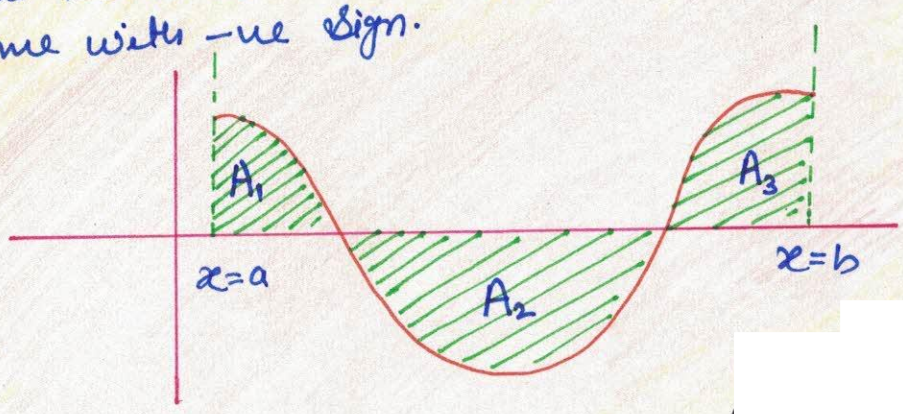

Definite Integration
Definite
Integration

DEFINITE INTEGRATION

Definition-1 $\int_a^b f(x)\,dx$ represent the algebrical sum of the area bounded by $y=f(x)$ with x-axis b/w the lines $x=a$ and $x=b$. Here, algebrical sum means, the area which is above x-axis will come with +ve sign and that below x-axis will come with -ve sign.



$$\int_a^b f(x)\,dx = A_1 - A_2 + A_3$$

$$\int_{-4}^{3} (3x-1)\,dx = \left[\frac{3x^2}{2} - x\right]_{-4}^{3}$$

$$\frac{3(9)}{2} - 3 - \left(\frac{3(16)}{2} + 4\right)$$

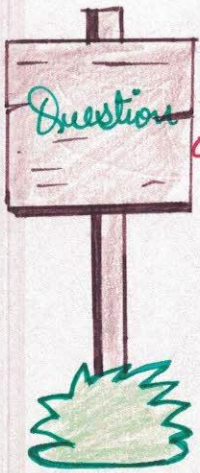
$$\frac{27}{2} - 3 - 24 - 4$$

$$= 13.5 - 31 = -17.5$$

PTR

1

If $\int_a^b f(x)\,dx = 0$, then it implies that the graph of $f(x)$ will cut x-axis atleast one time b/w (a,b). But, its vice versa is not true.



If $2a+3b+6c=0$, Then Prove that atleast one root of the quadratic eqn. $ax^2+bx+c=0$ lies in $(0,1)$.

$ax^2+bx+c=0$ lies in $(0,1)$

$$y\Big|_{x=0} = e\left[\frac{ax^3}{3}+\frac{bx^2}{2}+cx\right]_0^1$$

$$\int_0^1 f(x) = 0 \qquad \frac{a}{3}+\frac{b}{2}+c = z$$

$$2a+3b+6c = 6z$$

$$z=0$$

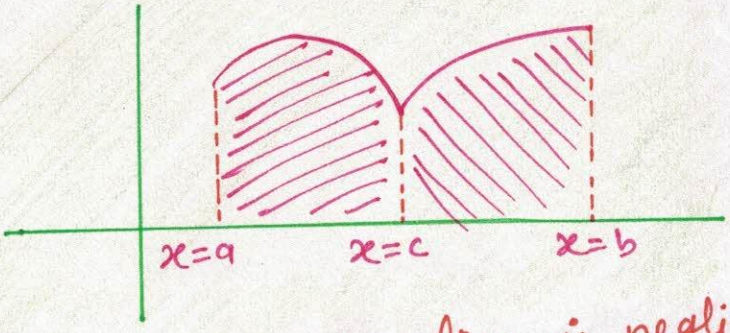
Area = 0

$\Rightarrow$ one root lies b/w $(0,1)$

2

If $f(x)$ is discontinuous at $x=c$. where $c \in (a,b)$

Then $\int_a^b f(x)\cdot dx$ will be equal to $= \int_a^c f(x)\cdot dx + \int_{+c}^b f(x)\,dx$



3

In DI area of a line is negligible which implies $\int f(x)$ in $[a,b]$, $[a,b)$, $[a,b]$ or (a,b) in same.

4

If $f(x)>0$ in (a,b), then $\int_a^b f(x)\cdot dx > 0$ Similarly, if $f(x)<0$ in (a,b), then $\int_a^b f(x)\cdot dx < 0$.

5. If $f(x)$ is above x-axis, then area is taken as the +ve and if $f(x)$ is below x-axis, then it is taken as -ive.

6. $$\int_a^b f(x)\,dx \le \left|\int_a^b f(x)\cdot dx\right| \le \int_a^b |f(x)|\cdot dx.$$

III. Equality holds when $f(x)$ lies completely above x-axis.

7. $$\int_a^b f(x)\cdot d\,g(x) = \int_{g^{-1}(a)}^{g^{-1}(b)} f(x)\cdot g'(x)\cdot dx$$

Question $\int_{-1}^{1} x^2 \cdot d(\ln x)$

$\ln x = -1 \quad x = \frac{1}{e}$ $\qquad$ $\ln x = 1 \quad x = e$

$$\int_{1/e}^{e} x^x \cdot \frac{1}{x}\cdot dx = \left[x^2/2\right]_{1/e}^{e} = \frac{e^2}{2} - \frac{1}{e^2 2}$$

$-x^2 = t^2 - 1$

Question $\int_0^{1/2} \frac{dx}{(1-2x)\sqrt{1-x^2}}$ $\qquad x = 1/t$

$1 - x^2 = t^2$

$-2x\,dx = 2t\,dt$

$dx = -\frac{1}{t^2}\cdot dt$

$$\int \frac{dx}{\frac{1}{t}\cdot\frac{1}{t^2}(t^2-2)\sqrt{t^2-1}} = \int \frac{t\,dt}{(t^2-2)\sqrt{t^2-1}}$$

$t^2 - 1 = u^2 \qquad 2t\,dt = 2u\,du$

$$\int \frac{u\,du}{(u^2-1)u} = \int \frac{du}{u^2-1}$$

$$= \frac{1}{2}\ln\left[\frac{u-1}{u+1}\right]$$

$$= \left[\frac{1}{2}\ln\frac{\sqrt{-x^2}-1}{\sqrt{-x^2}+1}\right]_0^{1/2}$$

8

$$\int_a^b \frac{d}{dx}(f(x)) \cdot dx = \int_a^b f^2(x) \cdot dx = |f(x)|_a^b$$

$$= f(b) - f(a)$$

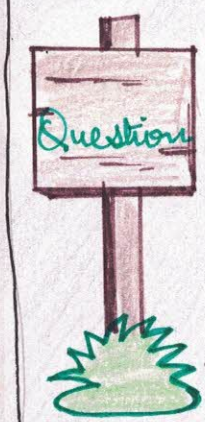
provide $f(x)$ is continuous in $[a, b]$.



$$\int_{-1}^{1} \frac{d}{dx}\left(\cot\left(\frac{-1}{x^2}\right)\right) \cdot dx$$

$$= \int_{-1}^{1} \frac{-1}{1+\frac{1}{x^2}} \left(-1/x^2\right) \cdot dx = \int_{-1}^{1} \frac{1}{1+x^2} \cdot dx$$

$$= \left[\tan^{-1} x\right]_{-1}^{1} = \frac{\pi}{4} - \left(\frac{-\pi}{4}\right) = \boxed{\frac{\pi}{2}}$$



$$\int_{-1}^{1} \frac{d}{dx}\left(\cot^{-1}\left(\frac{1}{x}\right)\right) \cdot dx$$

$$\int_{-1}^{0^-} \frac{d}{dx}\left(\cot^{-1}\frac{1}{x}\right) \cdot dx + \int_{0^+}^{1} \frac{d}{dx} \cot^{+}\left(1/x\right) \cdot dx$$

$$= \cot^{-1} 1/x \Big|_{-1}^{0^+} + \cot^{-1}\frac{1}{x}\Big|_{0^+}^{1}$$

$$= \pi - \frac{3\pi}{4} + \frac{\pi}{4} - 0$$

$$= \pi - \frac{\pi}{2} = \boxed{\frac{\pi}{2}}$$

9

$$\lim_{x\to\infty} \int_a^b f_n(x) \cdot dx = \int_a^b \lim_{x\to\infty} f_n(x) \cdot dx$$

Question

$$\lim_{x\to\infty} \int_{-a^{1/3}}^{a^{1/3}} \left(1-\frac{t^3}{n}\right)^n \cdot t^2 \cdot dt = \frac{2\sqrt{2}}{3} \quad \text{find } a.$$

$$\lim_{x\to\infty} \left(1-\frac{t^3}{n}\right)^n \cdot 0 = e^{-t^3}$$

$$I = \int_{-a^{1/3}}^{a^{1/3}} e^{-t^3} t^2 dt$$

$$-t^3 = 4$$

$$-3t^2 dt = dy$$

$$I = \int_{-a^{1/3}}^{a^{1/3}} e^y \cdot \frac{dy}{3} \qquad I = -\frac{1}{3} \cdot e^y \Big|_{-a^{1/3}}^{a^{1/3}}$$

$$I = -\frac{1}{3}\left[e^a - e^{-a}\right]$$

$$-\frac{1}{3}\left[e^a - e^{-a}\right] = \frac{2\sqrt{2}}{3}$$

$$-\left(\frac{e^{2a}-1}{e^a}\right) = 2\sqrt{2}$$

$$z^2 - 1 = 2\sqrt{2}\,z$$

$$z^2 + 2\sqrt{2}\,z - 1 = 0$$

$$z = \frac{-2\sqrt{2} \pm \sqrt{8+4}}{2} \Rightarrow \boxed{z = -\sqrt{2} \pm \sqrt{3}}$$

10

if $g(x)$ is inverse of $f(x)$ and domain of $f(x)$ is $[a,b]$ such that $f(a)=c$ and $f(b)=d$, then

$$\int_a^b f(x)\cdot dx + \int_c^d g(x)\cdot dx = bd - ac$$

As $g(x)$ is inverse of $f(x) \Rightarrow g(f(x)) = x$

$$f(x)\cdot x \Big|_a^b - \int_a^b x \cdot f'(x)\cdot dx + \int_c^d g(x)\cdot dx$$

$$\int_a^b g(f(x)) \cdot f'(x) \cdot dx$$

$f(x) = t$

$f'(x) \cdot dx = dt$

$$\int_c^d g(t) \cdot dt$$

$$x \cdot f(x)\Big|_a^b - \int_c^d g(t) \cdot dt + \int_c^d g(x)\, dx$$

$= f(b) \cdot b - f(a) \cdot a$

$= bd - ac$

Question

$$\int_0^1 e^{\sqrt{e^x}} \cdot dx + \int_e^{e^{\sqrt{e}}} 2 \ln(\ln x) \cdot dx$$

$e^{\sqrt{e}}$

$$\int_0^{\pi/2} \sin x \cdot dx \quad \text{or} \quad \int_0^{\pi/2} \cos x \cdot dx = 1$$

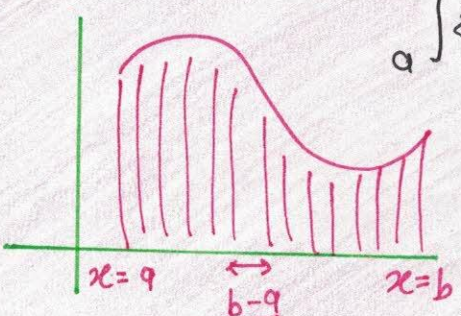
$$\int_0^{\pi/4} \sin^2 x \cdot dx = \int_0^{\pi/2} \cos^2 x \cdot dx = \pi/4$$

$$\int_0^{\pi/2} \sin^3 x\, dx = \int_0^{\pi/2} \cos^3 x\, dx = 2/3$$

$$\int_0^{\pi/2} \sin^4 x\, dx = \int_0^{\pi/2} \cos^4 x\, dx = 3\pi/16$$

DEFINITION 2 ...

$$\int_a^b f(x) \cdot dx = \lim_{x \to \infty} \sum_{r=1}^{n} \left(\frac{b-a}{n}\right) f\left(a + \frac{(b-a)r}{n}\right)$$



$a + \frac{b-a}{n}$

Area of the strip

$$\lim_{x \to \infty} \left[\left(\frac{b-a}{n}\right) f\left(a\left(\frac{b-a}{n}\right)r\right]\right.$$

$$\Rightarrow \int_a^b f(x) \cdot dx = \lim_{x \to 0} \sum_{r=1}^{n} \left(\frac{b-a}{n}\right) f\left(a + \frac{(b-a)r}{n}\right)$$

Question

$$\int_{-4}^{3} (3x-1)\,dx :\longrightarrow \lim_{x\to\infty} \sum_{r=1}^{n} \frac{7}{n} f\left(-4+\frac{7r}{n}\right)$$

$$\Rightarrow \lim_{x\to\infty} \sum_{r=1}^{n} \frac{7}{n}\left[3\left(-4+\frac{7r}{n}\right)-1\right]$$

$$\Rightarrow \lim_{x\to\infty} \sum_{r=1}^{n} \frac{7}{n}\left(-13+\frac{21r}{n}\right)$$

$$= \sum_{r=1}^{n}\left(-\frac{91}{n}\right) + \sum \frac{147r}{n^2}$$

$$= \lim_{x\to\infty}\left[\frac{-91}{n}\times n + \frac{147}{n^2}\left(\frac{n^2+n}{2}\right)\right]$$

$$= -\frac{91+147}{2} \qquad = -\frac{35}{2}$$

DEFINITION 3 ...

$$\int f(x)\cdot dx = g(x)+C \quad = \int_a^b f(x)\cdot dx = g(b)-g(a)$$

Variable of $\int$ in D.I is dummy variable.

$$\int_a^b f(x) = L = \int_a^b (t)\,dt = \int_a^b f(y)\cdot dy = \int_a^b f(k)\cdot dk$$

$$\int_a^b f(x)\cdot dx = -\int_b^a f(x)\,dx$$

if we use substitution rule in D.I for x Then it is must that limit of $\int$ will b change by substitution rule and Dont bring back the old variable.

NOTE

The Substitution of Variable will be in such a way that it has to be continuous within the limit of integration.

$$\int_a^b f(x)\cdot dx = \int_{-b}^{-a} f(-x)\cdot dx$$

$-x = t \Rightarrow -dx = dt$ $\quad \int_b^a f(t)\,dt = \int_a^b f(x)\cdot dx$

$$\int_a^b f(x)\cdot dx = \int_{a-c}^{b-c} f(x+c)\cdot dx$$

Question $\int_{26}^{5} \frac{2x}{1+x^2}\cdot dx$ $\Rightarrow$ $1+x^2 = t$ $\quad 2x\cdot dx = dt$

$$\int_{5}^{26} \frac{dt}{t} = \boxed{\ln\frac{26}{5}}$$

$$\int_a^b f(x)\,dx + \int_a^b g(x)\cdot dx + \int_a^b h(x)\cdot dx = \int_a^b (fx + gx + hx)\,dx$$

Here we have to keep in mind that limit of $\int$ in each D.I must be same.

If $f(x)$ is piece cosx continuous f^n $[a,b]$ where $c_1, c_2, c_3 \ldots\ldots c_n \in [a,b]$ are point of discontinuity.

$$\int_a^b f(x)\,dx = \int_a^{c_1} fx\,dx + \int_{c_1}^{c_2} f(x)\cdot dx \ldots\ldots \int_{cx}^{b} f(x)\cdot dx$$

Question

$$\int_0^2 [x^2-x+1]\,dx$$

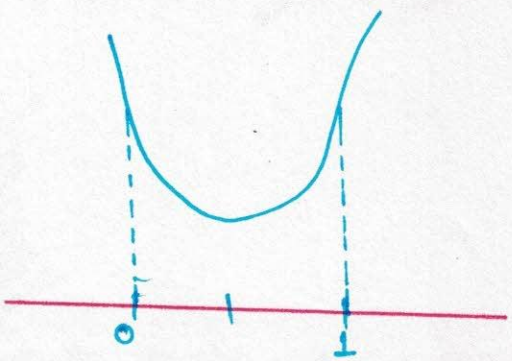
$x^2-x+1=1$

$x=0 \quad x=1$

$x^2-x+1=2$

$x=\frac{1+\sqrt{5}}{2}$ $\qquad x=\frac{1-\sqrt{5}}{2}$

$$\int_0^1 0\cdot dx + \int_1^{\frac{1+\sqrt{5}}{2}} 1\cdot dx + \int_{\frac{1+\sqrt{5}}{2}}^2 2\cdot dx$$



$$\frac{1+\sqrt{5}}{2} - 1 + 2\left(2-\left(1-\frac{\sqrt{5}}{2}\right)\right)$$

$$3-\left(\frac{1+\sqrt{5}}{2}\right)$$

$$\frac{6-(1+\sqrt{5})}{2} = \boxed{\frac{5-\sqrt{5}}{2}}$$

Question

$$\int_{-1}^{1} \Big[x+[x+[x+[x]]]\Big]\cdot dx$$

$[x+k]=[x]+k = \qquad k\in I$

$$\int_{-1}^{1} [x+[x+[x]+[x]]]\cdot dx$$

$$\Rightarrow \int_{-1}^{1} [x+[x]+[x]+[x]]\cdot dx$$

$$\Rightarrow \int_{-1}^{1} [x]+[x]+[x]+[x]\cdot dx$$

$$\Rightarrow 4\int_{-1}^{1} [x]\cdot dx \quad \Rightarrow 4\left(\int_{-1}^{0} -1\cdot dx + \int_{0}^{1} 0\cdot dx\right)$$

$$\Rightarrow -4$$

Question $\int_0^{5050\pi} [\tan^{-1} x] \cdot dx$

$\tan 1$ is the point where $[\tan^{-1} x]$ changes from 0 to 1.

$$\int_0^{\tan 1} 0 \cdot dx + \int_{\tan 1}^{5050\pi} 1 \cdot dx \Rightarrow 5050\pi - \tan 1$$

$0 - \frac{\pi}{2}$ we $[\tan^{-1} x]$ discont. $\quad x = \tan 1$

$$\int_1^{10\pi} [\sec^{-1} x] + [\cot^{-1} x] \cdot dx$$

$$\int_1^{10\pi} [\sec^{-1} x] \cdot dx + \int_1^{10\pi} [\cot^{-1} x] \, dx$$

$$\int_1^{\sec 1} 0 \cdot dx + \int_{\sec 1}^{10\pi} 1 \cdot dx + \int_1^{10\pi} 0 \cdot dx$$

$\Rightarrow 10\pi - \sec 1$

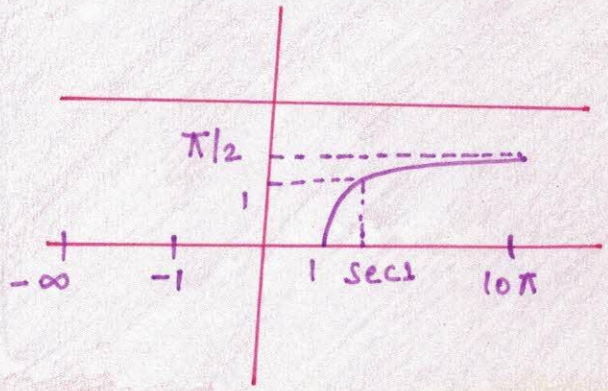
$\pi = 3.14$

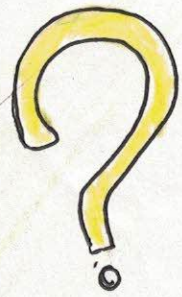
$[0, \pi]$

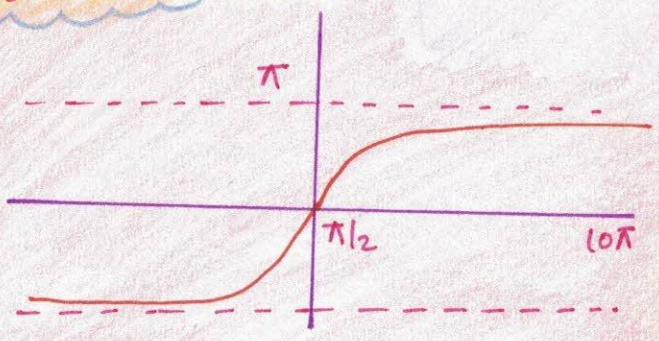
1, 2, 3

$[0, \pi]$

2, 3







KING'S FORMULA / GOD FORMULA

$$\int_a^b f(x) \cdot dx = \int_a^b f(a+b-x) \cdot dx$$

PROOF

$a+b-x=t$

$-dx=dt$

$$\int_b^a -f(t)\cdot dt = \int_a^b f(t)\cdot d \Rightarrow \int_a^b f(x)\cdot dx$$

$$I=\int_0^{\pi/2}\frac{\sin^3 x\,dx}{\sin x+\cos x} \quad \text{use king} \quad =\int_0^{\pi/2}\frac{\sin^3(\pi/2-x)\cdot dx}{\sin(\pi/2-x)+\cos(\pi/2-x)}$$

$$I=\int_0^{\pi/2}\frac{\cos^3 x}{\cos x+\sin x}\,dx$$

$$\text{king}+\text{add}=\int_0^{\pi/2}\frac{\sin^3 x+\cos^3 x}{(\sin x+\cos x)}\cdot dx$$

$$=\int_0^{\pi/2}\frac{\sin x+\cos x\left(1-\frac{2\sin x\cos x}{2}\right)\cdot dx}{\sin x+\cos x}$$

$$=\left[x+\frac{\cos 2x}{4}\right]_0^{\pi/2}=\frac{\pi}{2}-\frac{1}{4}+\frac{1}{4}=\frac{\pi}{2}$$

$$\int_{\pi/6}^{\pi/3}\sin 2x\,\ln(\tan x)\cdot dx=\int_{\pi/6}^{\pi/3}\sin 2\left(\frac{\pi}{2}-2x\right)\ln\left(\tan\left(\frac{\pi}{2}-2x\right)\right)dx$$

$$\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi+2\pi}{6}=\frac{\pi}{2}$$

$$=\int_{\pi/6}^{\pi/3}\sin 2x\cdot\ln(\cot x)\cdot dx$$

$$2I=\int_{\pi/6}^{\pi/2}\sin 2x\,(\ln 1) \Rightarrow \boxed{I=0}$$

$$\int_{50}^{100}\frac{\ln x}{\ln(x)+\ln(150-x)}$$

$$= \int_{50}^{100} \frac{\ln(150-x)}{\ln(150-x)} = \int 1 - \frac{\ln(150-x)\cdot dx}{\ln x + \ln(150-x)}$$

$$I = \int_{50}^{100} \frac{\ln(150-x)}{\ln(150-x)+\ln x}$$

$$2I = \int_{50}^{100} \frac{\ln x + \ln(150-x)}{\ln(150-x)+\ln x}$$

$$2I = \int_{50}^{100} \frac{\ln((150-x)\cdot x)}{\ln(150-x)\cdot x)} \cdot dx$$

$$2I = 50$$

$$\boxed{I = 2I}$$

$$\int_0^{\pi} \frac{dx}{1+2^{\tan x}} \Rightarrow I = \int_0^{\pi} \frac{dx}{1+2^{-\tan x}}$$

$$2I = \int_0^{\pi} \frac{2^{\tan x}+1}{2^{\tan x}+1} \cdot dx$$

$$2\pi = \pi$$

$$\boxed{\pi = \pi/2}$$



$$\int_0^{\pi/2} \frac{x \sin x \cdot \cos x}{\sin^4 x + \cos^4 x}$$

$$2\left(\pi/2 - x\right)$$

$$\Rightarrow \pi - 2x$$

$$I = \frac{1}{2} \int_0^{\pi/2} \frac{x \sin 2x}{\sin^4 x + \cos^4 x} \cdot d$$

$$(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x$$

$$1 - \frac{\sin^2 2x}{2}$$

$$\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} \cdot dx$$

$$I \Rightarrow \int \frac{(\pi/2 - x) \sin x \cos x}{\sin^4 x + \cos^4 x}$$

$$2I = \int_0^{\pi/2} \frac{\sin x \cos x [x + \pi/2 - x]}{\sin^4 x + \cos^4 x}$$

$$2I = \frac{\pi}{4} \int_0^{\pi/2} \frac{\sin 2x}{1 - \frac{1}{2}\sin^2 2x} \cdot dx$$

$$2I = \frac{\pi}{2} \int \frac{\tan x \sec^2 x}{\tan^4 x + 1} \Rightarrow \frac{\pi}{2} \int_0^{\pi/2} \frac{dt}{2(t^2+1)}$$

$$I = \int_0^{\pi/2} \frac{\pi}{8} \tan^{-1} \tan^2 x$$

$$\int_0^1 \cot^{-1}(1 - x + x^2)\, dx \Rightarrow I = \int (\cot^{-1}(1 - (1-x) + (1-x)^2)$$

$$I = \int_0^1 \cot^{-1}(x + x^2 - x + 1) \cdot dx$$

$$2I = \int \cot^{-1}(1 - x + x^2)\, dx + \pi/2 - \tan^{-1}(1 - x + x^2)\, dx$$

$$\int_0^1 \tan^{-1}\left(\frac{1}{x^2 - x + 1}\right) \cdot dx \Rightarrow \int_0^1 \tan^{-1}\left(\frac{x - (x-1)}{1 + x(x-1)}\right) \cdot dx$$

$$\Rightarrow \int_0^1 \tan^{-1}(x) - \tan^{-1}(x-1) \cdot dx$$

$$\int_0^1 \tan^{-1}(x) - \int_0^1 \tan^{-1}(1 - x + x) \cdot dx$$

$$I = \int_0^1 \tan^{-1}x + \int_0^1 \tan^{-1}x \cdot dx$$

$$I = 2\int_0^1 \tan^{-1}x\,dx = 2\left[\tan^{-1}x \cdot x - \frac{1}{2}\int \frac{2x}{x^2+1}\cdot dx\right]_0^1$$

$$I = 2\left[x\tan^{-1}x - \frac{1}{2}\ln(x^2+1)\right]_0^1$$

$$I = 2\left[\pi/4 - \frac{\ln 2}{2}\right] \Rightarrow \boxed{I = \pi/2 - \ln 2}$$



$$\int_0^{\pi/4} \frac{x}{1+\cos 2x+\sin 2x}\cdot dx$$

$$\int_0^{\pi/4} \frac{(\pi/4 - x)}{1+\cos(\pi/2-2x)+\sin(\pi/2-2x)} = \int_0^{\pi/4} \frac{\pi/4 - x}{1+\sin 2x+\cos 2x}$$

$\tan x = t$

$$I = \frac{\pi}{8}\int \frac{1+t^2}{1+t^2+2t+1-t^2}\,dx \qquad 2I = \int \frac{\pi/4}{1+\sin 2x+\cos 2x}\cdot dx$$

$$= \frac{\pi}{8}\int \frac{dt}{t+1} = \left[\frac{\pi}{16}\ln(\tan x)\right]_0^{\pi/4}$$



If $f(x)$ is an even f^n, then evaluate -

$$\int_{-2}^{2}\left(x^3 f(x) + x f''(x) + 2\right)\cdot dx$$

$$\int_{-2}^{2} x^3 f(x)\cdot dx + \int_{-2}^{2} x f''(x)\cdot dx + [2x]_{-2}^{2}$$

$$I = \int_{-2}^{2} (-x^3 f(x) - x f''(x) + 2)\cdot dx$$

$$2I = [4x]_{-2}^{2}$$

$$\boxed{I = 2[4] = 8}$$

NOTE

$$\boxed{\int_0^{\infty} \frac{\ln x \cdot dx}{ax^2+bx+a} = 0}$$

Here, Coeff of x^2 and constant must be Same.

Question

$$\int_0^{\infty} \frac{\ln x}{x^2+2x+4}\cdot dx$$

$x = 2t$

$dx = 2dt$

$$\int \frac{\ln(2t)\, 2\cdot dt}{4t^2+4t+4} = \frac{2}{4}\int \frac{\ln(2t)\,dt}{t^2+t+1}$$

$$= \frac{1}{2}\int \frac{\ln 2\, dt}{t^2+t+1} + \frac{1}{2}\int \frac{\ln t\, dt}{t^2+t^2+1}$$

$$= \frac{1}{2}\ln \int \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{\ln 2}{2}\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}t}{2}\right)\Big]_0^{\infty}$$

$$\boxed{= \frac{\ln 2}{\sqrt{3}}\cdot\frac{\pi}{2}}$$

PROPERTY 7

$$\int_a^b x \cdot f(x) \cdot dx = \frac{a+b}{2} \int_a^b f(x) \cdot dx$$

Provided $f(a+b-x) = f(x)$

$$I \int_a^b x f(x) \cdot dx$$

King $I = \int_a^b (a+b-x) \cdot f(a+b-x) \cdot dx$

$$2I = \int_a^b x f(x) + \left((a+b)-x\right) f(x)\, dx$$

$$I = \frac{a+b}{2} \int_a^b f(x) \cdot dx$$

$$I \; a+b \int_a^b f(x)\, dx$$



If $I_1 = \int_0^\pi x \cdot f(\sin^3 x + \cos^2 x \cdot dx$

$I_2 \int_0^\pi f(\sin^3 x \cos^2 x \cdot dx$

Then find the value of $\frac{I_1}{I_2} = ?$

$$I_1 = \frac{\pi}{2} \int f(\sin^3 x + \cos^2 x) \cdot dx$$

PROPERTY 8

$$\int_{-a}^{a} f(x) \cdot dx = \int_0^a f(x) \cdot dx + \int_0^a f(-x) \cdot dx$$

$$\int_a^0 f(x) \cdot dx = \int_{-a}^0 f(x) \cdot dx + \int_0^a f(x) \cdot dx$$

$x = -t \qquad dx = -dt$

$$I = \int_a^0 -f(-t)\,dt + \int_0^a f(x')\,dx$$

$$\Rightarrow \int_0^a f(-x)\cdot dx + \int_0^a f(x)\cdot dx$$

This property is useful only when limit of Definite Integration are symmetrical.

PROPERTY-9

$$\int_{-a}^{a} f(x)\cdot dx = \begin{cases} 0 & \text{when } f(x) \text{ is odd} \\ 2\int_0^a f(x)\cdot dx & \text{when } f(x) \text{ is even} \end{cases}$$

$$\int_{-3}^{3} \frac{x^2 \sin x}{x^4 + x^2 + \cos x}\cdot dx = 0$$

{Limit symmetrical, odd fn}

QUESTIONS ON PROPERTY

Question $\int_0^4 [x]\cdot dx = ?$

$$\int_0^1 0\cdot dx + \int_1^2 1\cdot dx + \int_2^3 2\cdot dx + \int_3^4 3\cdot dx$$

$$= (2-1) + 2(3-2) + 3(4-3) = 6$$

Question $\int_0^2 [x^2]\,dx = ?$

$$\int_0^1 0 \cdot dx + \int_1^{\sqrt{2}} 1 \cdot dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 \cdot dx + \int_{\sqrt{3}}^{2} 3 \cdot dx$$

$$= \sqrt{2} - 1 + (\sqrt{3} - \sqrt{2}) \cdot 2 + 3(-\sqrt{3} + 2)$$

$$= 5 - \sqrt{2} - \sqrt{3}$$



$$\int_0^{3/2} x[x^2] \cdot dx = ?$$

$$\int_0^1 0 \cdot x\,dx + \int_1^{\sqrt{2}} x \cdot dx + \int_2^{3/2} 2x \cdot dx$$

$$= \frac{x^2}{2}\Big|_1^{\sqrt{2}} + \frac{2x^2}{2}\Big|_{\sqrt{2}}^{3/2} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$



$$\int_0^3 |5x - 9| \cdot dx \; ?$$

$$\int_0^{9/5} (9 - 5x) \cdot dx + \int_{9/5}^{3} (5x - 9) \cdot dx$$

PROPERTY - 10

QUEEN'S FORMULA

$$\int_0^a f(x) \cdot dx = \begin{cases} 0 & f(a-x) = -f(x) \\ 2\int_0^{a/2} f(x) \cdot dx & f(a-x) = f(x) \end{cases}$$

$$I = \int_0^a f(a-x) \cdot dx$$

CASE-I

$$f(a-x) = -f(x)$$

$$2I = \int_0^a f(x) + f(a-x)\cdot dx$$

$$2I \int_0^a f(x) - f(x)\, dx$$

$$I = 0$$

CASE-II

$$f(a-x) = f(x)$$

$$I = \int_0^a f(x)\cdot dx$$

$$I = \int_0^{a/2} f(x)\cdot dx + \int_{a/2}^a f(x)\cdot dx \qquad x = a-t, \quad dx = -dt$$

$$I = \int_0^{a/2} f(x)\cdot dx + \int_{a/2}^0 -f(a-t)\cdot dt$$

$$I = \int_0^{a/2} f(x)\cdot dx + \int_0^{a/2} f(a-x)\cdot dx$$

$$I = \int_0^{a/2} f(x)\cdot dx + \int_0^{a/2} f(x)\cdot dx = 2\int_0^{a/2} f(x)\cdot dx$$

Working Rule of Definite Integration if the integrand is in the form of $\int_0^a f(x)\cdot dx$ Then Check.

$f(a-x) = -f(x)$ → if yes → $= 0$

$f(a-x) = f(x)$ → if yes → $2\int_0^{a/2} f(x)\cdot dx$

NO

Check whether $\int_0^a f(x)\cdot dx$ is easy to solve or $\int_0^a f(a-x)\cdot dx$ is easy.

Otherwise

Use king + Add

Question $\int_0^{2\pi} |\sin x|\cdot dx$

$$f(2\pi - x) = |\sin(2\pi - x)| = |\sin x| = f(x)$$

$$I = 2\int_0^{\pi} |\sin x|\cdot dx$$

$$f(\pi - x) = |\sin(\pi - x)| = |\sin x| = f(x)$$

$$I = 4\int_0^{\pi/2} |\sin x|\cdot dx$$

$$f(\pi/2 - x) = |\sin(\pi/2 - x)| = |\cos x|$$

$$I = 4\int_0^{\pi/2} \sin x\cdot dx = -4\left[\cos x\right]_0^{\pi/2} = 4$$

Question

$$\int_0^{2\pi} \cos^5 x \, dx = 0$$

$$2\int_0^{2\pi} \cos^5(2\pi - x) \cdot dx$$

$$= 2\int_0^{\pi} \cos^5 x \, dx = 0$$

Question

$$\int_0^{2\pi} \frac{x \cdot \sin^4 x}{\sin^4 x + \cos^4 x} \cdot dx$$

$$2I = \int_0^{2\pi} \frac{x \sin^4 x + (2\pi - x)\sin^4 x}{\sin^4 x + \cos^4 x} \cdot dx$$

$$2I = \int_0^{2\pi} \frac{2\pi \sin^4 x}{\cos^4 x + \sin^4 x} \cdot dx$$

$$I = \pi \int_0^{\pi} \frac{\sin^4 x}{\cos^4 x + \sin^4 x} \cdot d$$

$$I = 2\pi \int_0^{\pi} \frac{\sin^4 x}{\cos^4 x + \sin^4 x} \cdot dx$$

$$I = 4\pi \int_0^{\pi/2} \frac{\sin^4 x}{\cos^4 x + \sin^4 x} \cdot dx$$

King + Add

$$2I = 4\pi \int_0^{\pi/2} \frac{\sin^4 x + \cos^4 x}{\cos^4 x + \sin^4 x} \cdot d$$

$$I = 2\pi \int_0^{\pi/2} dx = 2\pi \left(\pi/2\right)$$

$$I = \pi^2$$

Question

$$\int_0^{\pi/2} \ln(\sin x)\cdot dx$$

Using + Add

$$2I = \int_0^{\pi/2} (\ln \sin 2x - \ln 2)\cdot dx$$

$$2I = \int_0^{\pi/2} \ln(\sin 2x)\cdot dx - \int_0^{\pi/2} \ln 2 \cdot dx$$

$$2I = \frac{2}{2}\int_0^{\pi/2} \ln \sin t \; dt - \pi/2 \ln 2$$

$$2I = I - \pi/2 \ln 2$$

$$I = -\pi/2 \ln 2$$

NOTE

- $\int_0^{\pi/2} \ln(\sin x)\cdot dx = \int_0^{\pi/2} \ln(\cos x)\cdot dx = -\frac{\pi}{2}\ln 2$
- $\int_0^{\pi/2} \ln(\tan x)\cdot dx = \int_0^{\pi/2} \ln(\cot x)\cdot dx = 0$
- $\int_0^{\pi/2} \ln(\sec x)\cdot dx = \int_0^{\pi/2} \ln(\operatorname{cosec} x)\cdot dx = \frac{\pi}{2}\ln 2$

$$\int_{-1}^{1} \frac{dx}{(1+x^2)(1+e^x)}$$

$$= \int_0^1 \frac{dx}{(1+x^2)(1+e^x)} + \int_0^1 \frac{e^x\cdot dx}{(1+x^2)(1+e^x)}$$

$$= \int_0^1 \frac{e^x + 1}{(1+x^2)(1+e^x)} \cdot dx$$

$$= \int_0^1 \frac{1}{1+x^2} \cdot dx = \tan^{-1} x = \frac{\pi}{4}$$

$$\int_0^1 \frac{\sin^{-1} x}{x} \cdot dx$$

$x = \sin\theta \qquad \theta \in (0, \pi/2)$

$$\int_0^{\pi/2} \frac{\theta \cos\theta}{\sin\theta} d\theta = \int_0^{\pi/2} \theta \cot\theta \cdot d\theta$$

$$= \int_0^{\pi/2} \cot\theta \cdot d\theta - \int_0^{\pi/2} 1 \cdot \int \underbrace{\cot\theta}_{\ln\sin\theta} \cdot d\theta$$

$$= \left[\theta \ln|\sin\theta|\right]_0^{\pi/2} + \frac{\pi}{2}\ln 2$$

$$I = \frac{\pi}{2}\ln 2$$

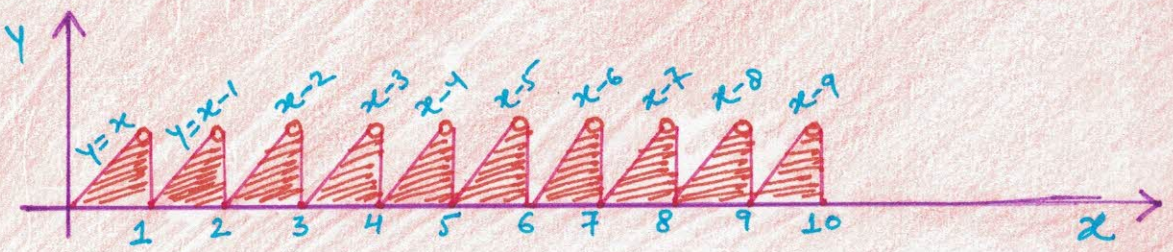
JACK FORMULA

$$\int_0^{nT} f(x) \cdot dx = n \int_0^{T} f(x) \cdot dx$$, where $f(x)$ is a periodic

f^n $x \in$ Integer with fundamental Period 'T'.

Question

$$\int_0^{10} \{x\} \cdot dx \qquad \int_0^{10.1} \{x\} \cdot dx = 10\int_0^1 \{x\} \cdot dx$$



$$= 10\int_0^1 x \cdot dx = 10\left[\frac{x^2}{2}\right]_0^1 = \frac{10^5}{2}[1] = 5$$

Question

$$\int_0^{1000} e^{x-[x]} \cdot dx$$

$$1000\int_0^1 e^{\{x\}} \cdot dx = 1000\int_0^1 e^x \cdot dx$$

$$= 1000\,(e^1 - e^0)$$

$$= 1000\,(e-1)$$

Question

$$\int_0^{200\pi} \frac{dx}{1+e^{\sin x}}$$

$$I = 100\int_0^{2\pi} \frac{dx}{e^{\sin x}+1}$$

$$2I = 100\int_0^{2\pi} \frac{e^{\sin x}+1}{e^{\sin x}+1} \cdot dx$$

$$2I = \frac{100\,(2\pi)}{1} \quad = \quad I = 100\pi$$

PROPERTY # 12

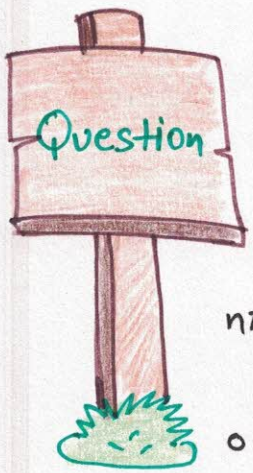
$$\int_a^{a+nT} f(x)\cdot dx = n\int_0^{T} f(x)\cdot dx$$

PROPERTY # 13

$$\int_{mT}^{nT} f(x)\cdot dx = (n-m)\int_0^{T} f(x)\cdot dx$$

PROPERTY # 14

$$\int_T^{T+a} f(x)\cdot dx = \int_0^{a} f(x)\cdot dx$$



$$\int_0^{n\pi+v} |\cos x|\cdot dx = ?$$

$$\frac{\pi}{2} < v < \pi$$
$$x \in N$$

$$\int_0^{n\pi} |\cos x|\,dx + \int_{n\pi}^{n\pi+v} |\cos x|\,dx$$

$$= n\int_0^{\pi} |\cos x|\,dx + \int_0^{v} |\cos x|\cdot dx$$

$$= 2n\int_0^{\pi/2} \cos x\,dx + \int_0^{\pi/2} \cos x\cdot dx + \int_{\pi/2}^{v} (-\cos x)\cdot dx$$

$$= 2n + 1 - [\sin x]_{\pi/2}^{v} \Rightarrow \boxed{2n + 2 - \sin v}$$



$$\int_0^{\pi} x(\sin^2\cdot\sin x + \cos^2\cos x)\cdot dx$$

$$2I = \int_0^{\pi} (\sin^2\sin x + \cos^2\cos x)\cdot dx$$

$$2I = 2\int_0^{\pi/2} \sin^2\sin x + \cos^2\cos x\cdot dx$$

$$I = \int_0^{\pi/2} \sin^2 x \cos x + \cos^2 x \sin x \cdot dx$$

$$2I = \int_0^{\pi/2} 2 \cdot dx$$

Question

If $f(x) + f(x+4) = f(x+2) + f(x+6)$

If $\int_x^{x+8} f(x) \cdot dx = g(x)$, then $g(4) = ?$

$x = x+2$

$$f(x+2) + f(x+6) - f(x+4) + f(x+8)$$

$$f(x+2) + f(x+6) = f(x) + f(x+4)$$

$$f(x+8) = f(x)$$

Periodic with Period $T = 8$

$$\int_0^8 f(x) \cdot dx = g(x) = \text{Const.}$$

NOTE

$\int_0^T f(x) \cdot dx = 0$ if $f(x)$ is an odd function.

NEWTON'S LEIBNITZ THEOREM

FORM I

$$y = \int_{f_1(x)}^{f_2(x)} g(x) \cdot dx$$

$$dy/dx = g(f_2(x)) \cdot f_2'(x) - g(f_1(x)) \cdot f_1'(x)$$

Question

$$f(x) = \int_3^{e^{3x}} \frac{t \cdot dt}{\ln t}$$

$$f'(x) = \frac{3e^{3x} \cdot e^{3x}}{\ln e^{3x}} - \frac{3}{\ln 3} \cdot 0$$

$$= \frac{3e^{3x}}{3x}$$

Question

$$x = \int_0^{y} \frac{dt}{\sqrt{1+4t^2}}$$

$$\frac{dx}{dy} = \frac{1}{\sqrt{1+4y^2}} \cdot 1$$

$$\frac{dy}{dx} = \sqrt{1+4y^2}$$

$$\frac{d^2y}{dx^2} = \frac{1}{2\sqrt{1+4y^2}} \cdot 8y \cdot \frac{dy}{dx} = \frac{1}{2\sqrt{1+4y^2}} \cdot 8y \cdot \sqrt{1+y^2}$$

$$\frac{d^2y}{dx^2} = 4y \qquad k = 4$$

Question

$$\int_0^{2\pi} \max\{\sin x, \sin^{-1}\cdot \sin x\} \cdot dx$$

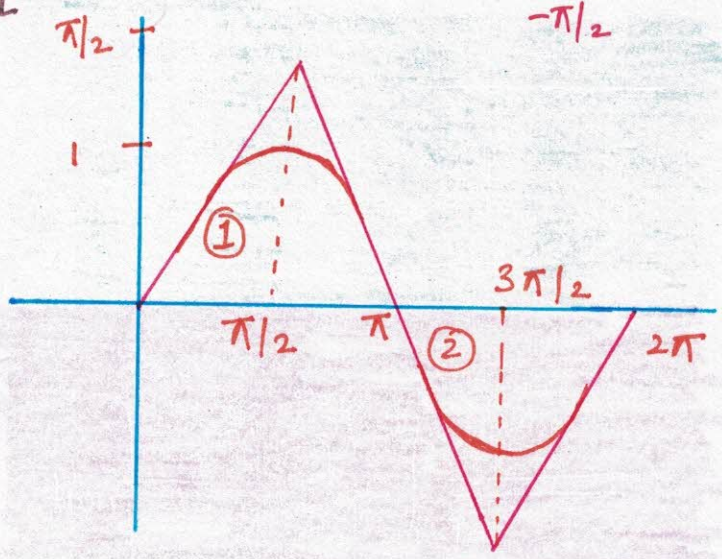
Question

Let $f(x)$ is continuous $f(x) > 0 \ \forall \ x \geq 0$ if

$$[f(x)]^{101} = 1 + \int_0^{x} f(t) \cdot dt \cdot f(101)^{100} = ?$$

Question Find max. value of f

$$f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(x)) \cdot dt$$



$$\text{area } ① = \frac{1}{2} \times \pi \times \pi/2$$

$$= \frac{\pi^2}{4}$$

$$\text{area } ② = \int_0^{\pi} \sin x \cdot dx$$

$$= \left[\cos x\right]_0^{\pi}$$

$$= -1 - 1 = -2$$

$$1 - n\left[\frac{\pi^2}{4} - 2\right]$$

Question

$$f(x)^{101} = 1 + \int_0^x f(t) \cdot dt$$

$$101 \cdot (f(x))^{100} \cdot f'(x) = f(x)$$

$$\int 101 (f(x))^{99} \cdot f'(x) \cdot dx = \int 1 \cdot dx$$

$$\frac{101 (f(x))^{100}}{100} = x + C$$

$f(0) = 1$

$\Rightarrow C = 1$

$$\frac{f(10)^{100}}{100} = \frac{x}{101} + C$$

$$f(101)^{100} = 100 + C = 100 + 1 = 101$$

$$f(x) = \sin x + \int_{-\pi/2}^{\pi/2} \sin x \cdot dt + \int_{-\pi/2}^{\pi/2} t f(t) \cdot dt$$

$$= \sin x + (\sin x)\pi + \int_{-\pi/2}^{\pi/2} t f(t) \cdot dt$$

(I) (II)

$$f(x) = \sin x (1+\pi) + A$$

$$A = \int_{-\pi/2}^{\pi/2} t \cdot f(t) \cdot dt$$

$$A = \int_{-\pi/2}^{\pi/2} t \cdot (\sin t (1+\pi) + A) \cdot dt$$

$$A = \int_{-\pi/2}^{\pi/2} t \sin t (1+\pi) \cdot dt + \int_{-\pi/2}^{\pi/2} At \cdot dt$$

$$A = (1+\pi) \int_{-\pi/2}^{\pi/2} t \sin t \, dt$$

$$A = 2(1+\pi) \int_{0}^{\pi/2} t \sin t \cdot dt$$

$$A = 2(1+\pi) (\sin t - t \cos t) \Big|_0^{\pi/2}$$

$$A = 2(1+\pi)$$

$$f(x) = (1+\pi) \sin x + 2(1+\pi)$$

$$f(x) = (1+\pi)(\sin x + 2)$$

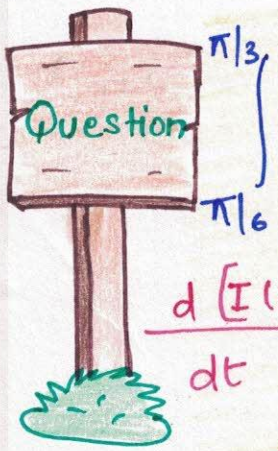
$$f(x)|_{max} = 3(1+\pi), \quad f(x)|_{min} = (1+\pi)$$

FORM - II

$\int_a^b f(x,t)\cdot dx = I(t)$ where a and b limits of x and t is a parameter independent of x,

Then $\frac{d[I(t)]}{dt} = \int_a^b f'(x,t)\,dx$.

Here $f'(x,t)$ = diff of $f(x,t)$ w.r.t. 't' considering x as a constant.



$$\int_{\pi/6}^{\pi/3} \sin(tx)\cdot dx = I(x,t)$$

$$\frac{d[I(x,t)]}{dt} = \int_{\pi/6}^{\pi/3} x\cos(tx)\cdot dx$$

$$= \left[\frac{x\sin tx}{t} - \frac{1}{t}\int \sin tx\cdot dx\right]_{\pi/6}^{\pi/3}$$

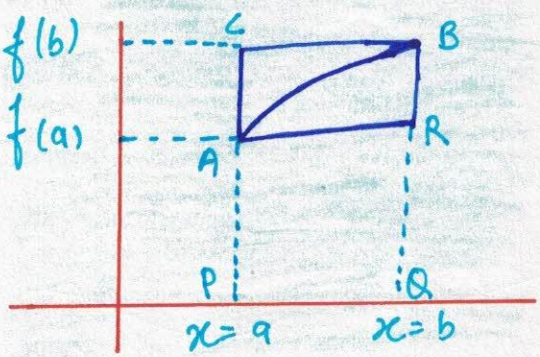
$$= \left[\frac{x\sin tx}{t} + \frac{\cos tx}{t^2}\right]_{\pi/6}^{\pi/3}$$

ESTIMATION PROPERTIES

CASE-I

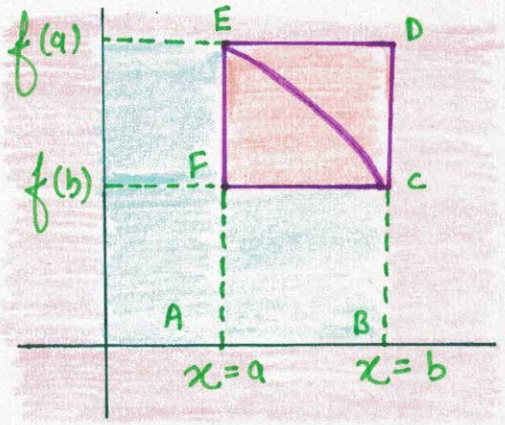
If $f(x)$ is monotonically increasing fn in $[a,b]$, then-

$$(b-a)f(a) \le \int_a^b f(x)\cdot dx < (b-a)f(b)$$



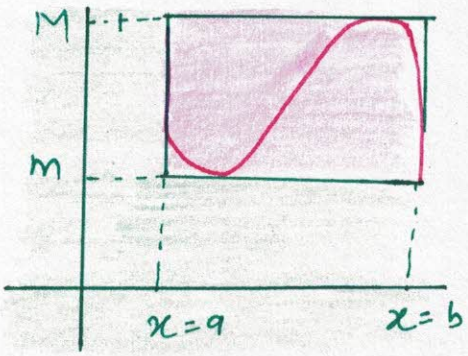
CASE-II If $f(x)$ is monotonically decreasing fn in $[a, b]$

Then -



$$(b-a)f(b) < \int_a^b f(x)\,dx < (b-a)f(a)$$

CASE-III



$$(b-a)m < \int_a^b f(x)\cdot dx < (b-a)M$$

m → min of $f(x)$ in $[a, b)$

M → max. of $f(x)$ in $[a, b]$

Question Prove that $\int_{\pi/4}^{\pi/2} (\sin x)^{10}\cdot dx$ is b/w $\pi/128$ to $\pi/4$.

$$f(\pi/4) = \left(\frac{1}{\sqrt{2}}\right)^{10} = \left(\frac{1}{2}\right)^5 = \frac{1}{16\times 2}$$

$$f(a) = \left(\frac{\pi}{4}\right)\left(\frac{1}{16}\right)\times\frac{1}{2} = \frac{\pi}{128}$$

$$a - b = \pi/4$$

$$f(b) = 1$$

$$\frac{\pi}{128} < f(x) < \pi/4$$

$\int_0^{\pi/2} \frac{\sin x}{x}\cdot dx$ $[-1, 1]$ $(0, \pi]$ $\frac{x\cos x - \sin x}{x^2}$ Decreasing fn

$$f(\pi/2) = \frac{2}{\pi} \qquad f(0) = 1$$

$$\frac{2}{\pi}\cdot\frac{\pi}{2} < f(x) < 1\cdot\pi/2$$

$$1 < f(x) < \pi/2$$

WALLI'S FORMULA ...

m and n are integers.

$$\int_0^{\pi/2} \sin^m x \cdot \cos^n x \cdot dx = \left[\frac{(n-1)(n-3)(n-5)\cdots 1 \text{ or } 2)(m-1)(m-3)\cdots 1 \text{ or } 2)}{(m+n)(m+n-2)(m+n-4)\cdots\cdots 1 \text{ or } 2}\right] \times K$$

$$K = \begin{cases} \pi/2 & \text{if } m, n \text{ both even} \\ 1 & \text{otherwise} \end{cases}$$

$$\int_0^{\pi/2} \sin^2 x \cos x \cdot dx = \frac{1 \cdot 1}{3 \cdot 1} \cdot 1 = \frac{1}{3}$$

$$\int_0^{\pi/2} \sin^7 x \cos x \cdot dx = \frac{(6)(4)(2)}{8 \cdot 6 \cdot 4 \cdot 2} \cdot 1 = \frac{1}{8}$$

$$\int_0^{\pi/2} \sin^4 x \cos^2 x \cdot dx = \frac{(3)(1) \cdot 1}{(6)(4)(2)} \cdot \frac{\pi}{2}$$

$$= \frac{\pi}{32}$$